

Table 3.4.1-5—Load Factors for Permanent Loads Due to Foundation Movements, γ_{SE}

Foundation Movement and Movement Estimation Method	SE
Immediate Settlement	
• Hough method	1.00
• Schmertmann method	1.40
• Local owner approved method	*
Consolidation settlement	1.00
Lateral Movement	
• Soil-structure interaction method (P-y or Strain Wedge)	1.00
• Local owner approved method	*
*To be determined by the owner based on local geologic conditions.	

Table 3.4.1-6—Load Factors for TU, γ_{TU}

	γ_{TU}
Displacements	1.20
Force effects	1.00

Where prestressed components are used in conjunction with steel girders, the force effects from the following sources shall be considered as construction loads, EL :

- In conjunction with longitudinal prestressing of a precast deck prior to making the deck sections composite with the girders, the friction between the precast deck sections and the steel girders.
- When longitudinal post-tensioning is performed after the deck becomes composite with the girders, the additional forces induced in the steel girders and shear connectors.
- The effects of differential creep and shrinkage of the concrete.
- The Poisson effect.

The load factor for live load in Extreme Event Load Combination I, γ_{EQ} , shall be determined on a project-specific basis.

Engineering judgment shall be exercised when applying blast loadings and when combining them with other loads.

3.4.2—Load Factors for Construction Loads

3.4.2.1—Evaluation at the Strength Limit State

All appropriate strength limit state load combinations in Table 3.4.1-1, modified as specified herein, shall be investigated.

When investigating Strength Load Combinations I and III for maximum force effects during construction, load

The most common applications of prestressed concrete in steel girder bridges are transverse post-tensioning of the deck and integral pier caps in which the tendons penetrate the girder webs. When a composite deck is prestressed longitudinally, the shear connectors transfer force to the steel. The effect of shrinkage and long-term creep around the shear connectors should be evaluated to ensure that the composite girder is able to recognize the prestressing over the life of the bridge. The contribution of long-term deformations in closure pours between precast deck panels which have been aged to reduce shrinkage and creep may need evaluation.

The Poisson effect recognizes the bulging of concrete when subjected to prestressing. When used in pier caps, post-tensioning causes a transverse Poisson tensile stress resulting in a longitudinal stress in the steel girders.

A load factor for passive lateral earth pressure is not given in Table 3.4.1-2 because, strictly speaking, passive lateral earth pressure is a resistance and not a load. For discussion of the selection of a passive lateral earth pressure resistance factor see Article 10.5.5.2.2.

Blast loads are considered an Extreme Event case of loading. However, not enough information exists at the time of this writing to determine what other loads should be combined with blast loads and the appropriate load factors.

C3.4.2.1

The load factors presented here should not relieve the contractor of responsibility for safety and damage control during construction.

Construction loads are loads that act on the structure only during construction. Often the construction loads are